



INTRODUCTION

“...It happened gradually, then suddenly...”

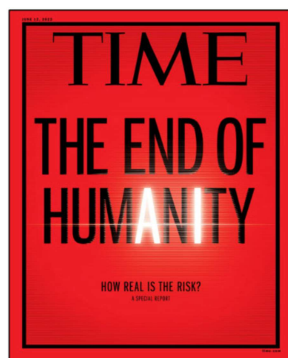
This famous quote from Ernest Hemingway’s 1926 novel, *The Sun Also Rises* could be used to describe some of the world’s most profound technological changes. Small advancements accumulate and then all of a sudden, the world is a different place.

That could describe the world before and after the birth of the internet and could now be applied to Artificial Intelligence (AI), which has ‘gradually, then suddenly’ burst onto the scene after nearly a century of research. It is almost impossible to browse the news or social media today, without seeing mention of Artificial Intelligence and the magazines Time, Science, Cosmopolitan and The Economist (to name only a few) have all dedicated cover stories to AI in 2023.



Cosmopolitan

www.cosmopolitan.com



Time

<https://time.com>



Economist

www.economist.com



Science

www.science.org

But where has AI come from and why is it suddenly such big news?

This report aims to answer those questions, by providing a general introduction to Artificial Intelligence technology, its business applications and its relevance to Travel & Tourism.

BRIEF HISTORY OF ARTIFICIAL INTELLIGENCE (AI)

AI has gained significant attention in recent years – and especially in 2023 – but AI is not new and can trace its history back to the development of computers after the Second World War, with the Dartmouth Conference in 1956 bringing together researchers from multiple fields to explore “thinking machines”. This is widely considered the start of AI as a distinct field of study and where the term “Artificial Intelligence” was used for the first time by the visionaries at that conference.

But it was not until the turn of the century that AI really came to the public’s attention, when the **IBM Deep Blue** supercomputer beat chess grandmaster Gary Kasparov in 1997, with artificial intelligence algorithms developed by IBM engineers. A few years later, in 2011, the **IBM Watson** computer won the US gameshow Jeopardy, after

being trained on a huge data set. These events showed the world for the very first time that computers powered with large amounts of data and artificial intelligence software – albeit very powerful computers out of reach of the general public - were capable of outperforming humans at very complex tasks.

Over the past decade several companies have taken this a step further and developed AI systems that have achieved incredible results and performed tasks not possible by humans, due to the huge scale and complexity of the challenge. **AlphaFold** from **Google Deepmind** is one example. The programme uses AI to predict the 3D structure of nearly every protein in the human body. This improves our understanding of diseases like Alzheimer's and can massively accelerate research into medicines, vaccines and drugs. Healthcare is therefore one of the most important and useful applications of AI.

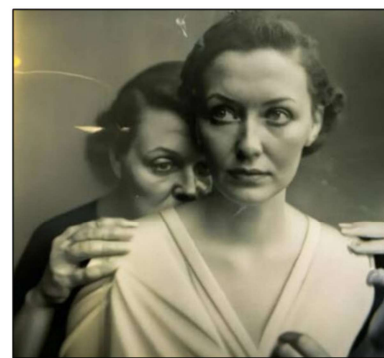
In 2018 a painting called “Edmond de Belamy, from La Famille de Belamy” was sold at Christie's auction house in New York for a staggering \$432,500 (USD). It was generated with artificial intelligence after being trained on many images of portraits from the 18th and 19th centuries and was the first piece of AI art sold at auction. The sale sparked a significant debate about the future role of humans in creative professions, such as art and digital media.



AI Painting (2018)
Edmond de Belamy



AI Painting (2023)
Girl with Glowing Earrings



AI Photo (2023)
Pseudomnesia : The Electrician

But this was only the start of a 'boom' in AI generated creative content. In 2023, a museum in The Hague (Netherlands) loaned the world famous, *Girl with Pearl Earring* (c. 1665) by Johannes Vermeer to an international exhibition and temporarily replaced it in their gallery with an AI inspired version entitled *Girl with Glowing Earrings*, with many visitors to the gallery believing it was a real painting. In April 2023 the German artist Boris Eldagsen won the 2023 Sony World Photography Award with an image entitled *Pseudomnesia: The Electrician*, before refusing the prize after revealing it was created by AI. (Pseudomnesia is Latin for fake memory!)

These images show how far AI creative technology has advanced in only a few short years, from an ability to generate low quality AI portraits in 2018, to high quality AI paintings and photorealistic AI images today. Both the *Girl with Pearl Earring* and *Pseudomnesia* images were created with AI software that is now available to everyone online and is partly why AI has taken off with the public in 2023, enabling any of us to become potential artists.

But paintings and photography are not the only form of art that has been shaped by AI. In 2023 the Grammys announced that music created with AI would be eligible for awards and Paul McCartney announced he had used AI to extract John Lennon's voice from an old demo tape. In late 2023, this track was released as the final Beatles Song *Now and Then*¹ which drew considerable public attention to AI across all generations. Canadian singer Grimes has also invited anyone to create new songs with an AI copy of her voice, offering to split the royalties equally on any AI track that is commercially successful.

Going a step further, an area of AI that is still in development is AI generated video. Imagine being able to produce a corporate marketing video easily and in only a few moments on your office laptop, or even turning your favourite book into a movie using AI. Whilst AI generated video does not currently exist at this advanced level yet, it is clear that AI is rapidly moving forward and has huge potential for both businesses and society.

But there are also risks and concerns about the widescale adoption of AI, which are covered later in this report. The use of AI has led to some highly publicised events, such as a strike by Hollywood actors and writers in 2023, partly due to concerns about AI's impact on the movie industry. Their concerns include the ability of AI to write scripts and the potential to create an AI digital replica of an actor (or extra), which could be included in films, without the physical actor present and potentially long after they have died. The full impact of AI on jobs of the future is yet to be determined, but it is possible that labour strikes will be experienced in other industries as AI rolls out further into society and other areas of work.

There is also a concern that AI could be used for disinformation (the deliberate sharing of false information), especially to interfere with political elections. High quality, photorealistic – but fake – images have already gone viral on the internet, such as an AI generated image of the Pope wearing a puffer jacket and an image purporting to show the arrest of former US President Donald Trump. Both images were generated by AI and both are fake, but they are photorealistic, very convincing, generated on a home computer in only a few minutes and were shared very widely across the internet and social media. It is therefore clear that the wide scale implementation of AI across society needs 'guardrails' for its appropriate use, which could range from formal regulations to codes of conduct and best practice guidance. These options are discussed further in an accompanying WTTC report on **'AI Risks & Governance'**.



Viral AI-generated (fake) images of the Pope and former US President Donald Trump

But not all celebrities are against AI and digital replicas – often known as 'deepfakes'. For example World Cup winning Argentinian footballer Lionel Messi has allowed a deepfake version of himself to advertise snacks, as part of his sponsorship deal with PepsiCo. At the website www.messimessages.com anyone can generate and download to their phone a short personalised deepfake video message from Messi who invites your friends by name, to bring food and watch the big game with him – and it can even get Messi to provide the message in his voice in English, Spanish, Portuguese or Turkish!



Argentinian footballer Lionel Messi : Deepfake Messi Messages

AI has therefore been regularly hitting the news headlines and as with all new trends, a few high profile cases can propel a technology that is initially only used by a few early adopters, into global mass adoption. For AI this happened with the international release of a sophisticated AI powered chatbot, named **ChatGPT**, in November 2022 from a company called OpenAI. This caught the public's attention, registering over 100 million users in only 2 months and making it (at the time) the fastest growing consumer product of all time. This tipped AI into the public mainstream in 2023 and sparked the current worldwide interest in AI.

While ChatGPT was the first AI chatbot to reach mass consumption, others are now available including **Google Bard**, **IBM Watsonx.ai** and **Microsoft Copilot**, with others produced around the world, such the **Ernie** AI chatbot from Baidu in China.

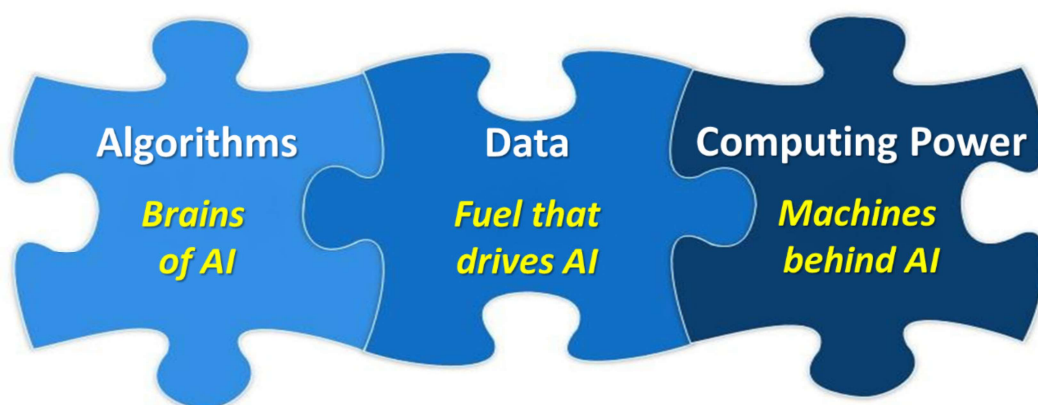
These sophisticated AI chatbots are not the only way to use AI, but they have exposed the power of AI to the general public, which everyone can now access on their desktop computers and smartphones (and through digital connections to the 'cloud'), unlike the 1990's when the use of AI was limited to a very small number of owners of supercomputers (such as IBM Deep Blue). The future of AI is exciting and evolving very fast. It is impossible to accurately predict how AI will shape our lives over the next 5 to 50 years, but the following section explains why now is a crucial moment for the future of AI and society.

WHAT IS ARTIFICIAL INTELLIGENCE & WHY THERE IS GLOBAL INTEREST NOW

One of the most important aspects of **AI is that it is a multi-use technology**. Like electricity it can be applied in lots of different ways, to lots of different scenarios.

There is no single, universally accepted definition for Artificial Intelligence, but the Oxford English Dictionary defines AI as “*the capacity of computers, or other machines, to exhibit intelligent behaviour*”. This means AI systems appear to think, learn and act like humans and in some cases exceed the capabilities of humans. AI systems can analyse vast amounts of data, solve complex problems, make decisions and perform creative tasks.

Some AI technologies have been around for more than 50 years, but advances in computing power, the availability of enormous quantities of data and new developments in software algorithms have led to major AI breakthroughs in recent years. It is these three components of **advanced algorithms**, **data** and **computing power**, that explain how machines can exhibit intelligent behaviour and why AI has suddenly exploded into our everyday lives.



*Algorithms tell computers what to do. Data tells computers what to learn.
Computing power gives machines the power to learn and make decisions*

ALGORITHMS : THE BRAINS OF AI

'Traditional programming' involves encoding human knowledge and experience into a set of precise rules that a computer can follow, step-by-step, which make the computer appear to respond intelligently.

These **rules, called algorithms, tell computers how to perform tasks** and in traditional programming are often expressed in an 'IF-THEN-ELSE' format, which resembles a decision tree. For example to create a 'digital doctor', an algorithm might look like:

```
IF the patient has a fever

THEN prescribe Drug X

ELSE send the patient home
```

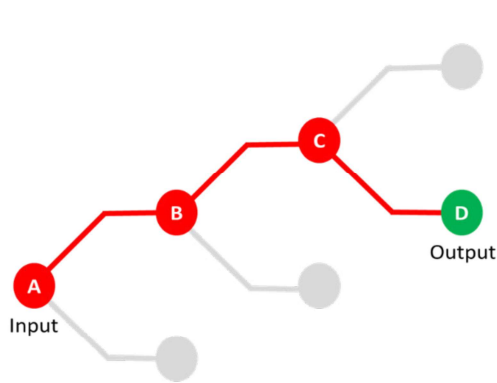
The intelligence in traditional computer systems comes directly from human knowledge and expertise being recorded into a format that a computer can process. Humans can read this 'computer code' and understand how these digital systems make specific decisions and can therefore identify mistakes and improve the system. For example, in the 'digital doctor' scenario, new clauses could be added to deal with special cases, or to reflect new medical knowledge. For example:

```
IF the patient has a fever AND is allergic to Drug X

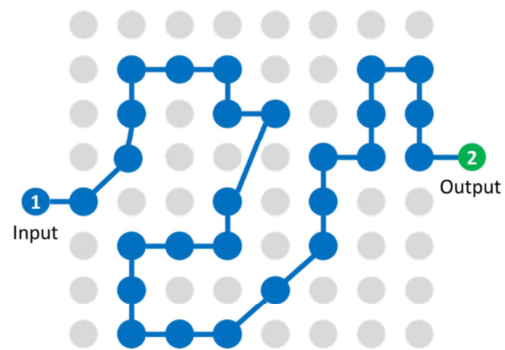
THEN prescribe Drug Y

ELSE prescribe Drug X
```

This example now starts to illustrate a key limitation of traditionally programmed systems. To develop a useful and reliable ‘digital doctor’, a huge number of rules and exceptions would be required that the system would very quickly become very large and very complicated. Thousands of IF-THEN-ELSE rules would still be unlikely to capture all of a real doctor’s expertise and experience gained over time. Therefore **traditional programming of ‘intelligent systems’ is best suited to constrained environments which do not change much over time and where the rules can be strictly defined**. A good example of this could be an intelligent ‘tax calculator’ which when presented with an individual’s financial income and other relevant circumstances could efficiently review a series of ‘IF-THEN-ELSE’ rules to calculate their tax liability.



Traditional Programming



Artificial Intelligence (AI)

Unlike traditional programming, **AI systems are not explicitly coded in a series of precise steps, but instead ‘learn’ how to solve a problem by examining training data**. In the example above, a traditional programme (**in red on the left**) goes through a series of defined steps and pre-programmed decision points at A, B and C to reach an outcome D (**in green**), whereas in the AI example (**in blue on the right**) the system calculates the best route to the outcome (the snake to point 2) based on the data it has been trained on (at point 1), not on explicitly defined and pre-programmed decision points. **AI systems can also continuously ‘learn from experience’** and therefore the route taken to the outcome might be different each time the AI system is run (like a car driver taking different routes through a city based on experience of traffic conditions at different times of day).

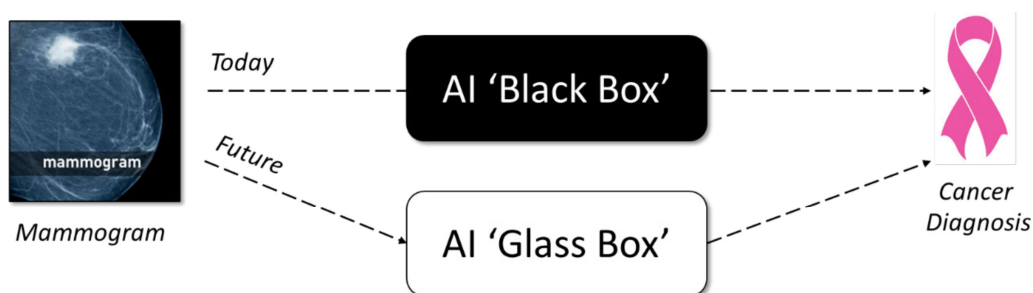
Returning to a ‘digital doctor’ example, an ‘AI digital doctor’ could be trained on thousands (or millions!) of medical scans from sick and well patients to learn how to detect tumours, without following pre-defined rules which are otherwise very difficult to codify and write. Many human doctors may say that experience plays a significant role in clinical diagnosis, but an AI system can be trained on far more x-ray images than a doctor could ever experience, leading to AI systems having remarkable capabilities in areas such as breast cancer detection. For example a Swedish AI study in 2023 found that AI-supported breast cancer screening (the analysis of mammogram images) was as good as two radiologists working together ². It also reduced their workload by 50% and led to 20% more cancers being detected early for treatment ³. This illustrates some of the incredible capabilities of AI and its benefit to society.

But this also leads to an important concept in AI called **‘Explainable AI’**. It is possible to verify the output of an AI system as correct (for example did a patient really have cancer or not), but it may not be possible to know exactly how the AI system got to a decision or outcome, as the AI has calculated the ‘interim steps’ (e.g. the blue snake in the diagram above) by itself, without human programming of the specific steps to be taken. **AI systems are therefore sometimes referred to as ‘black boxes’ with details of exactly how they work and operate - in some cases - not fully known, or understood**. This can even include their developers, as they too can often only verify the output and not the steps taken by the AI.

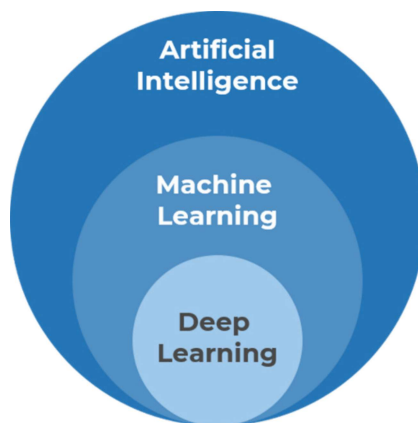


'Explainable AI' is therefore an active and ongoing field in which researchers are trying to understand and articulate how AI systems reach an output. This is crucial to **improve trust in artificial intelligence systems** (i.e. to understand the magic inside the black box and to turn a 'black box' into a 'glass box').

The 'digital doctor' example nicely illustrates why 'explainable AI' is so important (in this case for both the patient and the doctor) to understand why an AI system has made a decision and believes a patient has breast cancer before sending them for intrusive cancer treatment.

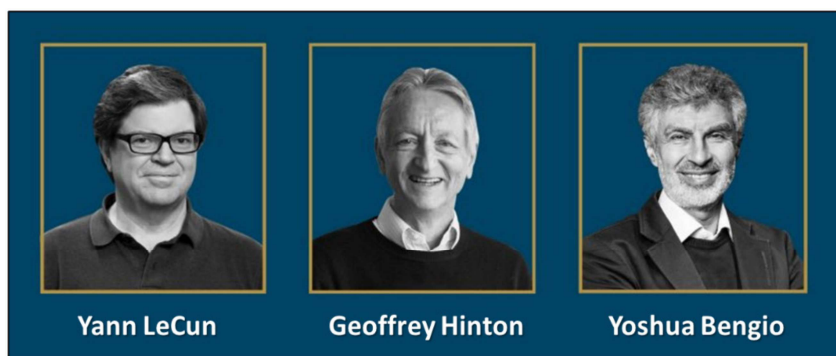


When considering how an AI system is constructed, there are **two primary methods for how an AI system 'learns' to perform a task**, which are machine learning and deep learning.



- **Machine Learning** : Machine Learning (ML) systems can learn from huge amounts of data and continuously improve their performance over time when provided with more and/or better quality training data. With this 'knowledge' gained from training data, machine learning powered AI systems can then make predictions (such as for weather forecasting), or recognise patterns in data (such as for image and speech recognition).
- **Deep Learning** : Deep Learning (DL) is a more sophisticated subset of Machine Learning and uses complex processes inspired by the human brain called Artificial Neural Networks (ANN). How neural networks function is beyond the scope of this 'Introduction to AI' report, but deep learning systems can excel at very complex tasks such as generating new works of art and medical drug discovery.

Researchers **Yann LeCun**, **Geoffrey Hinton** and **Yoshua Bengio** are credited with laying the foundations for how AI systems can 'learn'. Their pioneering research on deep learning in the 1990's and early 2000's led to major breakthroughs and their work is now used in a variety of AI powered applications, from self-driving cars to medical diagnosis. In 2018 the three researchers – nicknamed the '**Godfathers of AI**' – were awarded the **Turing Award**, which is considered the highest honour in computer science (equivalent to a Nobel Prize) for their fundamental contributions to artificial intelligence.



Media nicknamed "Godfathers of AI"¹⁴

The Turing Award is named after **Alan Turing** who was a brilliant British mathematician. He decoded the German Enigma machines during the Second World War (which were considered impossible to crack) by building one of the world's first computers and was one of the main architects of modern computer science following the war. He is often considered one of the 'founding fathers' of modern computing.

In 1950 Alan Turing published a groundbreaking paper called 'Computing Machinery and Intelligence' which introduced the '**Turing Test**' for artificial intelligence. This is still used today to test a machine's ability to exhibit intelligent behaviour and if it is indistinguishable from a human. At this time (early 2024) no computer has decidedly passed the Turing Test, but there have been some very convincing contenders. Modern, sophisticated, AI chatbots (such as Google Bard and Microsoft Copilot) can be very difficult to distinguish from a human. The Turing Test is also a useful way to measure the progress of AI research, and since 1950 has helped to raise international awareness of the potential for machines to think and behave like humans.

Whether an AI system uses machine learning, or deep learning techniques to '**learn**', there are different ways that AI algorithms can be '**trained**', but the most common approaches are:

- **Comparing AI-generated outcomes or predictions to the correct answer:** this is known as '**Supervised Learning**' in which AI models are trained on data that has been correctly labelled (for example an AI animal recognition system could be trained on images of animals correctly labelled as cats or dogs).
- **Finding patterns in data:** this is known as '**Unsupervised Learning**' in which AI models are trained on data that has not been labelled. In this case the AI system learns by finding patterns in the data (for example edges, shapes, features, or colours in images of animals that are common to cats or dogs).
- **Trial & Error:** this is known as '**Reinforcement Learning**' in which AI models 'learn by doing'. Each outcome of the AI training is either mathematically **rewarded** or **punished**, in a 'trial and error' feedback loop (for example an AI animal recognition system that identified a certain breed of dog as a cat would be 'punished', but 'rewarded' it if correctly identified it as a dog, even though it had not seen that breed before)

At the core of every AI system is a set of algorithms (often using machine, or deep learning) and AI training techniques (such as supervised, unsupervised or reinforcement learning) which determine how the system processes information, learns patterns and makes decisions. As new algorithms are developed artificial intelligence is becoming increasingly capable. But AI requires two other important components to be successful – data and computing power.

DATA : THE FUEL THAT DRIVES AI

“Data is the new oil” is a phrase often associated with the digital age. Data is the fuel behind modern computing and AI algorithms, allowing them to learn, find relationships in data and make informed predictions and decisions.

In today’s world, huge volumes of data are created as we go about our everyday lives. From text messages, emails, documents and social media posts, to photographs and videos on our smart phones. Humanity is generating massive amounts of information every day and even more data is created by millions of sensors around the world and in our homes, cars, cities, public transport infrastructure and factories.



An often quoted statistic from a 2013 study ⁵ is that **90% of the worlds data has been generated in the last 2 years**. That rate is likely to be even quicker today. A 2023 paper by the European Parliament on ‘*Artificial Intelligence Threats & Opportunities*’ ⁶ estimates that by **2025 the volume of data produced in the world each year could be 175 zettabytes** (up from less than 1 zettabyte a year generated in 2010, which is a 175 x increase in the volume of data in just 15 years). To put that in context, a top-of-the-range laptop in 2024 may come with a 1 terabyte hard drive, which is a billion times smaller than a zettabyte. Therefore the volume of global data produced in a single year in 2025, may be equivalent to 175 billion laptop computers....or 21 laptops for every person on earth!

Data	Bytes	Size
1 Kilobyte (KB)	1,000	10 ³
1 Megabyte (MB)	1,000,000	10 ⁶
1 Gigabyte (GB)	1,000,000,000	10 ⁹
1 Terabyte (TB)	1,000,000,000,000	10 ¹²
1 Petabyte (PB)	1,000,000,000,000,000	10 ¹⁵
1 Exabyte (EB)	1,000,000,000,000,000,000	10 ¹⁸
1 Zettabyte (ZB)	1,000,000,000,000,000,000,000	10 ²¹

The volume of digital data in the world is rising so fast, and getting so large, that in 2022 scientists formally agreed two new units of measurement for the first time in 31 years. These are the **Ronnabyte (RB)** and the **Quettabyte (QB)**. A Ronnabyte is a billion, billion, billion bytes (!) and a Quettabyte is a thousand Ronnabytes...or 1 followed by 30 zeros! Scientists believe this will be sufficient for at least the next 25 years, but the relentless pace of digital innovation means we may all be using Quettabytes quicker than anyone expected!

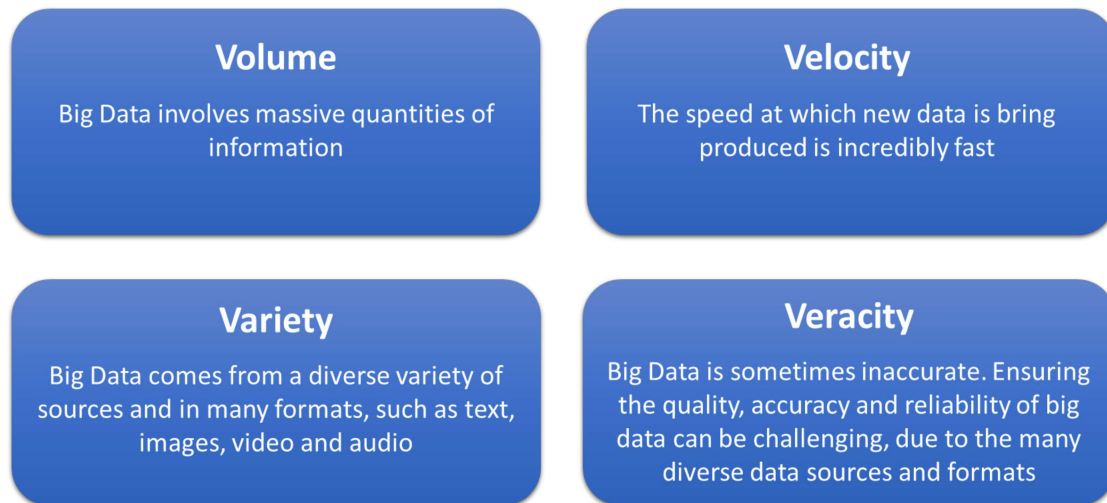
Data	Bytes	Size
1 Ronnabyte (RB)	1,000,000,000,000,000,000,000,000,000	10^{27}
1 Quettabyte (QB)	1,000,000,000,000,000,000,000,000,000,000	10^{30}

This data explosion is largely driven by three factors:

1. **Increasing numbers of smartphones and internet devices:** the number of devices connected to the internet grows every year. There are several estimates for the total number of networked devices in the world, but all agree that it is many billions of devices. The CISCO Annual Internet Report 2023 ⁷ estimates there are 29.3 billion networked devices in the world, which is more than 3 x the global population of 8 billion people (in 2023).
2. **Growth of social media:** social media platforms generate many terabytes of text, video and audio data every day (1 terabyte = approximately 1500 CD's) with billions of messages and posts made every minute. In 2020, during the height of the COVID-19 pandemic, the Head of WhatsApp (Will Catchcart) issued a tweet stating that approximately 100 billion messages were being exchanged every day on WhatsApp alone and it is estimated that videos shared on platforms such as TikTok and YouTube account for over half of all internet traffic. Social data is rising at an incredible pace, but is also very useful for AI systems to understand human behaviour.
3. **New data collection and storage technologies:** new technologies, such as sensors placed throughout our cities, energy grids and transport infrastructure, are generating huge amounts of data on human usage patterns. Similarly the '**Internet of Things (IoT)**', which are the networked everyday items (such as fitness trackers and Smart TV's) are growing in popularity and producing new streams of data. The CISCO Annual Internet Report 2023 estimates that of the 29.3 billion connected devices, 14.7 billion are IoT devices. Nearly 50% of these are connected home appliances (such as home security systems) and 30% are connected car applications (such as networked vehicle diagnostics tools, emergency calling systems and in car entertainment platforms).

This volume of data is also leading to a significant rise in **cloud computing and data storage centres**, with the U.S. hosting 10x more data centres than any other country in the world (in 2023) ⁸.

These extremely large and very complex datasets are often called '**Big Data**' and are characterised by their **volume, velocity, variety** and **veracity**, commonly referred to as the **4 V's**.



This **exponential rise in the quantity, availability, quality and diversity of data** around the globe is powering the current wave of AI technology. **Data provides the information required for AI models to learn and improve**, and is the fuel for AI systems to undertake very complex tasks at scale.

In particular 'big data' supports the rise of AI in the following ways :

1. **Training Data:** AI systems require substantial amounts of training data to learn patterns. This makes their predictions and decisions more accurate, which is especially important in areas such as self-driving cars where massive datasets of images and videos, from many different road scenarios, are captured and used to train AI algorithms, which can then perceive and navigate complex environments by themselves.
2. **Feature Extraction:** large and diverse datasets help AI algorithms to accurately identify and extract features from within data. This enables more accurate insights from AI systems, such as AI powered image recognition tools.
3. **Model Performance:** big data enhances the performance and robustness of AI models, allowing them to handle a wide range of situations and inputs. This is especially important in sectors such as healthcare, where AI systems can analyse extensive patient and medical data to assist in diagnosis, treatment and drug discovery.
4. **Predictive Analytics:** large datasets and AI can be used by businesses to forecast trends, customer behaviour and market shifts. This provides companies with actionable insights and helps with corporate decision making.
5. **Personalisation:** when AI systems can access large volumes of data on human behaviours, preferences, and interactions they can provide tailored suggestions and experiences for users. This is useful for e-commerce and social media platforms to suggest products, content, or services tailored to an individual user.
6. **Natural Language Processing (NLP):** NLP models, such as AI powered chatbots, virtual assistants and language translation systems, benefit from diverse and extensive language data, such as books, articles and social media posts. This allows an AI NLP model to understand and generate realistic human-like language very effectively.
7. **Real-Time Decision Making:** as the 'velocity' of big data accelerates, in areas such as banking transactions, AI systems can process and analyse this data in (near) real time to provide instant recommendations, or decisions, for functions such as fraud detection.